

Machine Learning Assignment 1

Polynomial Regression

Roll Number: IMT2024086

1 Introduction

The objective of this assignment is to develop polynomial regression models for two personalised datasets, Var1 and Var2, and use them to predict the continuous target variable y . Polynomial degree was selected empirically using cross-validation rather than assuming a fixed degree.

For each problem, all available input features were used. Polynomial features were generated for a range of degrees and evaluated using 5-fold shuffled cross-validation. Ordinary Least Squares (OLS), Ridge, Lasso, and Elastic Net regression were considered, with several regularisation strengths for the regularised models.

The main model-selection criterion was the mean cross-validated R^2 , while cross-validated MSE was also considered. The metrics are

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

and

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (2)$$

The polynomial features were standardised before regression. After the best degree and regression method were selected, the model was refitted using the complete training dataset and used to generate the test predictions.

2 Var1: Results and Analysis

Var1 contains six input features corresponding to operational parameters of the geothermal steam turbine. The target variable is the Net Power Score.

The cross-validation curves in Figure 1 show a large reduction in MSE as the polynomial degree increases from 1 to 5. The corresponding R^2 increases rapidly and reaches its maximum around degree 5. Beyond degree 5, the improvement becomes negligible and the MSE slightly increases. Therefore, degree 5 was selected empirically.

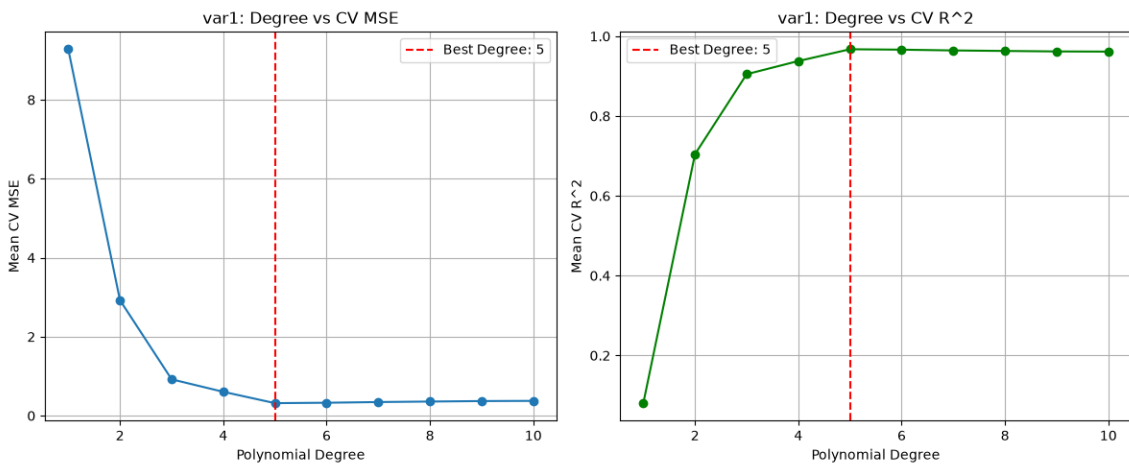


Figure 1: Var1 cross-validation MSE and R^2 versus polynomial degree.

The selected Var1 model is a **degree-5 Lasso regression** with $\alpha = 0.01$. Its mean cross-validation performance is

CV MSE	CV R^2
0.3174	0.9674

The residual diagnostics are shown in Figure 2. The residuals are distributed around the zero-error line without a strong systematic pattern. The residual histogram is approximately symmetric and centred close to zero. This indicates that the degree-5 model captures the dominant nonlinear relationship without an obvious major residual trend.

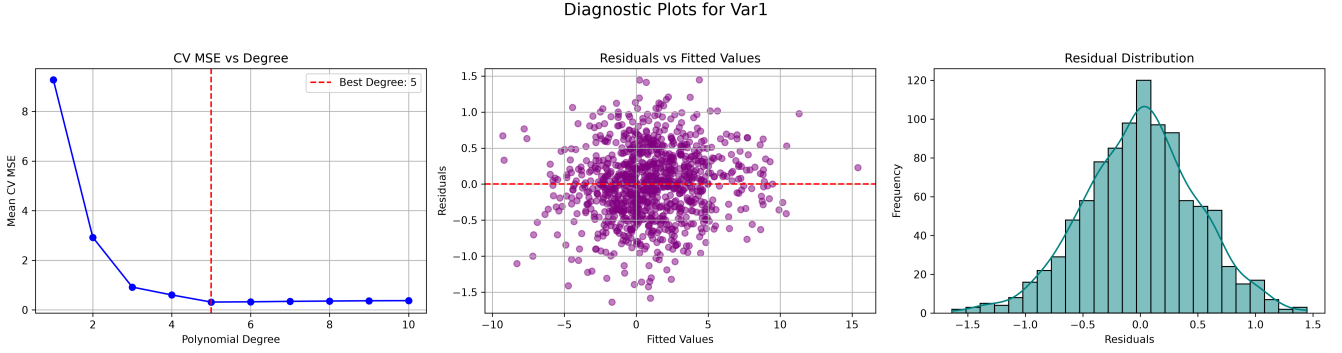


Figure 2: Diagnostic plots for the selected Var1 model.

The use of Lasso regularisation is useful because a degree-5 polynomial with six input features produces many polynomial and interaction terms. Regularisation helps control unnecessary coefficients and reduces the risk of overfitting.

3 Var2: Results and Analysis

Var2 contains three spatial coordinates representing the East-West, North-South, and vertical depth offsets. The target variable is the Thermal Anomaly Score.

The degree-selection curves in Figure 3 show a very large reduction in MSE during the initial increase in polynomial degree. The cross-validated R^2 rapidly approaches 1 and then begins to saturate. The empirical model-selection procedure identifies **degree 11** as the best degree.

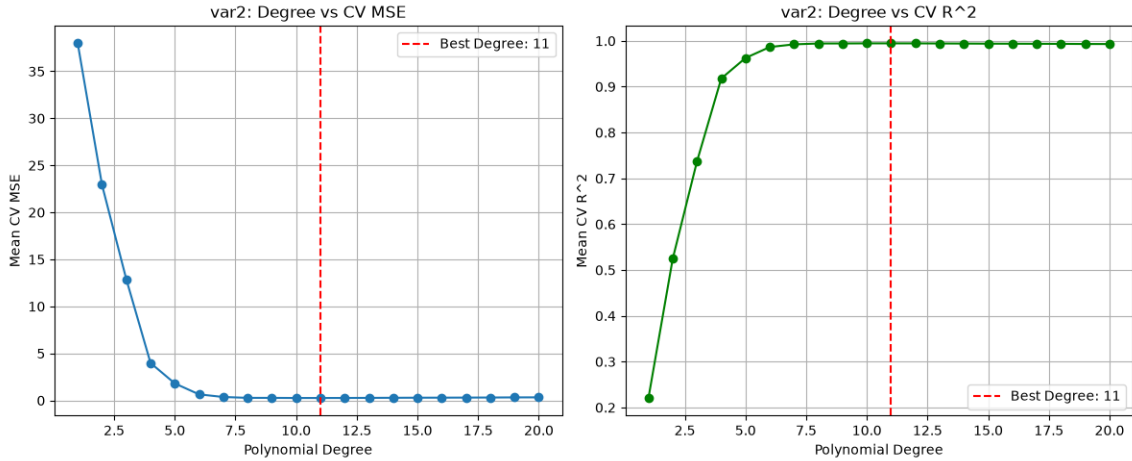


Figure 3: Var2 cross-validation MSE and R^2 versus polynomial degree.

The selected Var2 model is a **degree-11 Ridge regression** with $\alpha = 1.0$. Its mean cross-validation performance is

CV MSE	CV R^2
0.2573	0.9945

The residual diagnostics are shown in Figure 4. The residuals are concentrated around zero and do not show a strong systematic relationship with the fitted values. The residual distribution is approximately symmetric and centred near zero, which is consistent with the high cross-validated R^2 .

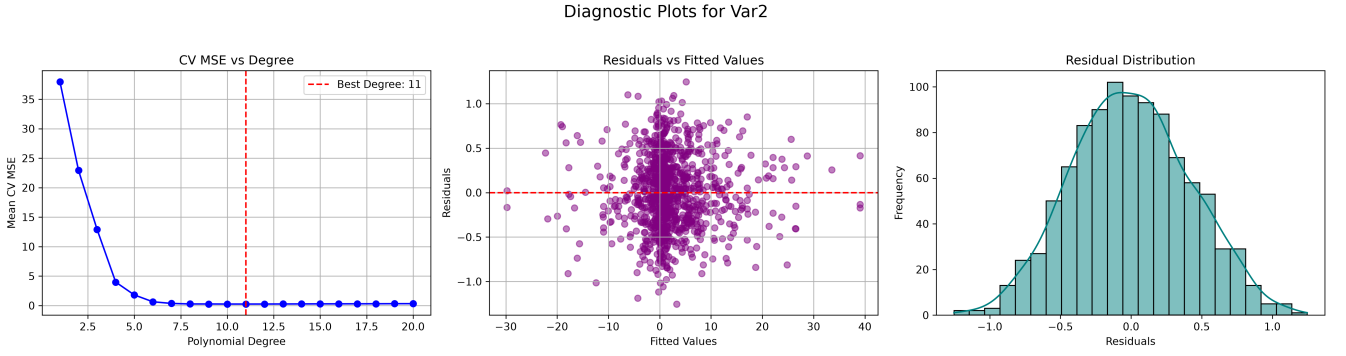


Figure 4: Diagnostic plots for the selected Var2 model.

The higher polynomial degree is appropriate for Var2 because its underlying relationship is more complex. Ridge regularisation helps control the large number of correlated polynomial terms generated by the degree-11 expansion.

4 Model Comparison

Table 1 summarises the final models.

Table 1: Final model configurations and cross-validation performance.

Property	Var1	Var2
Input features used	6	3
Selected degree	5	11
Regression method	Lasso	Ridge
α	0.01	1.0
CV folds	5	5
CV MSE	0.3174	0.2573
CV R^2	0.9674	0.9945

Var2 achieves the stronger cross-validation performance, with a R^2 of approximately 0.9945 compared with 0.9674 for Var1. Its lower MSE also indicates smaller prediction errors under cross-validation.

The selected degrees differ considerably between the two problems. Var1 reaches its best performance at degree 5, whereas Var2 benefits from a substantially higher degree of 11. This demonstrates why the polynomial degree should be determined from the data rather than fixed in advance.

The different regularisation methods are also appropriate for the respective models. Lasso was selected for Var1, while Ridge was selected for Var2. Both provide regularisation that helps control the complexity introduced by polynomial feature expansion.

5 Final Prediction and Conclusion

After model selection, each selected model was refitted on the complete training dataset. The resulting models were then used to generate predictions for the corresponding test datasets.

The final configurations are:

Problem	Final Model
Var1	Degree-5 Lasso, $\alpha = 0.01$
Var2	Degree-11 Ridge, $\alpha = 1.0$

Overall, polynomial regression successfully models both datasets. Var1 achieves strong predictive performance with a relatively moderate degree-5 polynomial, while Var2 requires a higher degree to capture its more complex nonlinear structure.

The residual plots for both problems show errors distributed around zero without a pronounced systematic pattern. Together with the high cross-validation R^2 values, this indicates that the selected models provide good representations of the relationships present in the training data.

The final models therefore provide a suitable balance between model complexity and predictive performance, with polynomial degree selected empirically using cross-validation and all available input features retained.

6 GitHub Repository

All files used for data preprocessing, model training, polynomial degree selection, cross-validation, evaluation, and test-set prediction are available in the following GitHub repository:

https://github.com/sujay-ravva/ML_Assignment_1